Active Knowledge Modeling: A Proactive Paradigm for User Alignment in Human-AI Collaboration

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Abstract

Many failures in human-AI collaboration do not begin at inference time. They begin earlier, when a system starts acting without an explicit model of the user it is serving. This creates repeated explanation cost, unstable personalization, and generic downstream output. We define **Active Knowledge Modeling (AKM)** as a proactive user-modeling paradigm in which AI guides and elicits the user's goals, constraints, preferences, background, and decision style, structures that knowledge explicitly, and injects it into the workflow before downstream tasks begin. AKM differs from passive user modeling, one-off prompt customization, and retrieval-only personalization because it treats user understanding as an explicit upstream layer rather than an emergent side effect. We position **Dialogue-as-Elicitation (DaE)** as the first complete implementation of AKM in persona and advisory collaboration [10]. AKM is the parent concept for proactive user modeling; DaE is its first complete reference implementation. Existing DaE research and SSRN assets should therefore be understood as derived research evidence and early public timestamps for the broader AKM paradigm, rather than as the canonical source of the concept itself. The central claim of this paper is simple: systems should model the user first, then execute tasks.

Keywords: active knowledge modeling, proactive user modeling, pre-task user modeling, user modeling, AI personalization, human-AI collaboration, context injection, persona profiling, dialogue as elicitation, reference implementation

1 Introduction

Personalized AI often fails in a familiar way. The system appears capable, but its outputs remain generic, unstable, and weakly fitted to the operator. The user then spends repeated effort re-explaining goals, constraints, preferences, and context. This is not only a prompting problem. It is an upstream modeling problem.

Recent work has described hidden labor in making AI usable as *alignment debt* [7]. Related work on personalization benchmarks has shown that large language models can benefit from richer user context [8, 4, 2]. Persona-oriented dialogue research also demonstrates that explicit representations

of speaker identity can materially affect outputs [6, 1]. Preference elicitation and interaction-structure work further shows that higher-quality outputs often depend on better elicitation rather than better decoding alone [3]. Prompt design research provides more systematic ways to shape local task behavior [11], while human-centered chatbot design work highlights the role of boundary conditions and design tensions in sensitive interaction settings [9]. Yet these strands still leave a structural gap: most systems begin task execution before they have built a stable and explicit model of the user.

This paper names that missing layer: **Active Knowledge Modeling (AKM)**. AKM is a proactive paradigm in which AI guides the elicitation of a user model, structures it explicitly, and injects it into downstream workflows before any substantive task work begins.

The paper makes three contributions. First, it defines AKM as a distinct upstream modeling paradigm rather than a variant of prompting or memory. Second, it differentiates AKM from passive user modeling, one-off prompt customization, and retrieval-only personalization. Third, it positions Dialogue-as-Elicitation (DaE) as the first complete implementation of AKM in persona and advisory collaboration, while clarifying that existing DaE papers and SSRN assets are derived research evidence, not the canonical source of AKM itself.

2 Problem Formulation

Most AI workflows assume the system can begin acting immediately and refine its understanding of the user later. This default is convenient, but structurally weak. Important user information is tacit, fragmented, and expensive to transfer through ad hoc interaction.

The result is predictable:

1. users repeatedly restate themselves;
2. models rely on inference and guesswork;
3. outputs drift toward generic competence instead of operator fit;
4. personalization remains shallow and unstable across tasks.

From a human-AI collaboration perspective, this also reflects a misalignment of incentives: the system optimizes for response generation before the user model has been made explicit [5]. The issue is therefore not merely that current systems answer imperfectly. It is that they often answer too early.

3 Definition of AKM

Active Knowledge Modeling (AKM) is a paradigm in which AI first guides and elicits a model of the user, then makes that model explicit and structured, and finally injects it into downstream workflows before task execution begins.

At minimum, the modeled information should include:

- goals,
- constraints,
- preferences,
- background,
- decision style.

In stronger implementations, the model should also surface tensions, trade-off logic, recurring failure modes, and evidence quality. The crucial move is not merely to store more user information. It is to elevate user understanding into a deliberate upstream layer.

3.1 Necessary Properties

For a workflow to qualify as AKM, three properties should hold.

1. **Guided elicitation:** the system actively helps uncover operator information rather than merely waiting for self-description.
2. **Explicit structure:** the resulting user model is represented in a durable, inspectable form.
3. **Pre-task injection:** the model is loaded into downstream work before substantive execution begins.

If one of these properties is missing, the workflow may still be useful, but it is no longer AKM in the strict sense used here.

4 What AKM Rejects

AKM should be distinguished from three older paths.

4.1 Passive User Modeling

Passive user modeling learns slowly from behavioral traces, long-term interaction, or accumulated memory. This can be useful, but it leaves the system guessing for too long and keeps the user model opaque.

4.2 One-Off Prompt Customization

One-off prompt customization can improve a single session, and prompt pattern design can make that work more systematic [11]. But it still does not create a stable, reusable model of the operator.

4.3 Retrieval-Only Personalization

Retrieval-only personalization gives the system access to notes, files, or prior fragments. However, information access is not the same as an explicit upstream user model. Without a designed user-model layer, downstream work still begins from incomplete operator understanding.

5 Why AKM Operates at a Higher Level

AKM does not ask, "How should this answer be improved?" It asks, "Has the system acquired a stable enough model of the user before answering at all?"

This distinction matters because many downstream failures are not caused by weak generation quality. They are caused by starting from the wrong user model, or from no user model at all. AKM addresses that upstream defect directly.

Conceptually, AKM shifts personalization from a reactive correction layer to a proactive conditioning layer. In that sense, AKM sits above task prompting and below domain execution: it prepares the operator model against which downstream prompts, tools, and agents should work.

6 Related Work and Positioning

AKM sits at the intersection of several adjacent lines of work without collapsing into any single one of them.

The first line is **persona-aware dialogue and fixed persona conditioning**. Work such as persona-based neural conversation models and fixed-persona language agents shows that explicit persona representations can shape downstream behavior [6, 1]. AKM agrees with that intuition, but shifts the emphasis from conditioned generation to the upstream construction of the operator model itself.

The second line is **personalization and user modeling**. Benchmarks and datasets for personalized reading and creative writing show that richer user-specific context improves relevance and fit [8, 4, 2]. AKM extends this line by treating personalization not as an optional downstream feature but as an explicit pre-task modeling layer.

The third line is **elicitation and interaction structure**. Preference elicitation research demonstrates that better questioning or interaction format can materially improve decision support [3]. AKM inherits that insight, but applies it to a broader user-model construction problem rather than to a narrow local preference query.

The fourth line is **human-AI alignment labor**. Alignment debt and related human-centered design work highlight the hidden effort users expend to make AI systems usable [7, 9]. AKM contributes a systems-level response: instead of paying that cost repeatedly through session-by-session correction, systems should convert it into an explicit reusable user-model asset.

Prompt engineering research remains relevant but insufficient. Prompt patterns can improve local

instructions [11]; AKM argues that better local prompting still does not replace an explicit upstream operator model.

7 Reference Implementation: DaE

Dialogue-as-Elicitation (DaE) is the first complete implementation of AKM in persona and advisory collaboration [10]. DaE uses structured questioning to elicit a reusable user profile before downstream planning, writing, coding, research, or advisory work begins.

The distinction between AKM and DaE should be kept strict:

- AKM is the parent paradigm;
- DaE is the first complete implementation in a high-value domain;
- existing DaE papers and repositories are derived research assets within the AKM lineage.

Without this distinction, DaE is misread as an isolated skill or product. With it, DaE becomes the first working proof that AKM can function as a reusable upstream layer.

8 Relationship to Existing Research Assets

Existing DaE papers, repositories, and SSRN assets should be understood as *derived research evidence* and public timestamps for the broader AKM paradigm. They are strategically useful because they provide early public traceability, implementation evidence, and bridge material for future attribution. They should not, however, be mistaken for the canonical source of AKM itself.

This distinction is important for both scholarship and strategy. Scholarly clarity requires separating the parent concept from a particular implementation. Strategic clarity requires ensuring that future citations, derivatives, and implementations can be organized under AKM rather than forcing all subsequent work back into the narrower DaE label.

9 Application Map

Although DaE is the first complete implementation, AKM is not limited to persona or advisory collaboration. The same paradigm can be extended to strategic advisory, learning, health management, style and presentation, and other domains where downstream quality depends heavily on stable understanding of the operator.

These are not separate conceptual inventions. They are application branches of the same upstream modeling paradigm.

10 Methodological Scope and Boundary Conditions

This paper is primarily conceptual. Its aim is not to claim that AKM has already been exhaustively validated across domains, nor to argue that every personalization workflow should be reclassified under AKM. The narrower claim is that a distinct upstream layer exists, that this layer has been undernamed in prior discussion, and that DaE provides an early complete implementation in one concrete domain.

Two boundary conditions follow.

First, AKM should not be confused with generic memory accumulation. A system that stores user fragments without active elicitation or structured pre-task injection does not satisfy the paradigm.

Second, AKM should not be treated as a replacement for downstream expertise. AKM improves the starting point for downstream work; it does not eliminate the need for domain-specific reasoning, tools, or safety constraints.

11 Implications for System Design

If AKM is taken seriously, then user alignment should be designed as infrastructure rather than ornament.

For product builders, this implies dedicated elicitation stages, inspectable user-model artifacts, and explicit loading of those artifacts into downstream workflows.

For researchers, it implies that evaluation should not stop at answer quality alone. It should also examine whether the user model itself is explicit, reusable, updateable, and causally responsible for downstream improvements.

For operators, it implies that the effort spent clarifying oneself can be converted from repeated session cost into reusable context capital.

12 Conclusion

AKM reframes user alignment as a pre-task modeling problem. Its central proposition is straightforward: **model the user first, then execute the task.**

If that proposition holds, then many systems currently described as personalized are still operating one layer too low. They optimize outputs without first stabilizing the user model that those outputs depend on.

AKM names that missing layer. DaE provides its first complete implementation. Existing SSRN and research assets provide early evidence and timestamped support. The strategic objective is therefore not merely to ship another product, but to establish AKM as the parent concept under which future implementations can be consistently understood.

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